

Offshore Stuck Pipe & Pack-off Prevention Study

Section: Mechanics of Mechanical Binding and Pack-Off

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Mechanical sticking and pack-off hazards in extended-reach drilling are predominantly driven by inefficient hole cleaning, cuttings bed accumulation, and reactive shale swelling.

In deep intervals (>3000m MD), inadequate annular velocity allows cuttings to build up on the low side of the hole. When pulling out of hole or rotating below 90 RPM, the BHA acts as a wedge, precipitating instantaneous pack-off and pressure spikes exceeding 400 psi.

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Section: Proactive Mitigation & Hydraulic Hole Cleaning Directives

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Drilling crews in reactive smectite shale must deploy tandem high-density viscous pills every 90 meters drilled.

Maintain drillstring rotation at 110-130 RPM during reaming and reciprocate pipe smoothly to agitate cuttings beds into high-velocity laminar flow channels.

If pump pressure surges unexpectedly with torque fluctuations, stop penetration immediately, pull bit off bottom by 3 to 5 meters, and circulate bottoms-up at maximum allowable flow rate before resuming drilling.